

Creating a Map

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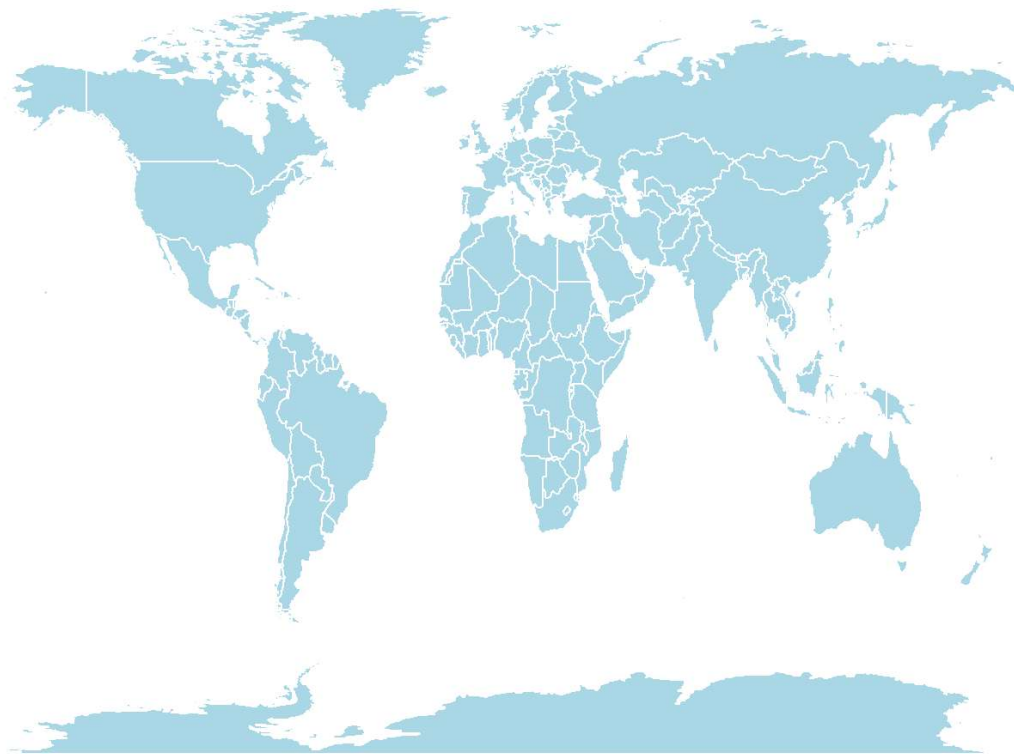
2025-05-07

```
library(pacman)
```

```
## Warning: package 'pacman' was built under R version 4.4.3
```

```
p_load(tidyverse)
```

```
map<- map_data("world")  
ggplot(map, aes(x=long, y=lat, group = group))+  
  geom_polygon(fill="lightblue", colour="white")+  
  theme_void()
```



```

North_Asia<- c("China", "Japan", "Mongolia", "North Korea", "South Korea", "Taiwan")
North_Asia_Map<- map_data("world", region = North_Asia)

region.data<- North_Asia_Map%>%
  group_by(region)%>%
  summarise(long=mean(long), lat=mean(lat))%>%
  arrange()

ggplot(North_Asia_Map, aes(x=long, y=lat))+
  geom_polygon(aes(group=group, fill=region))+
  geom_text(data=region.data, aes(label = region),
            size=5, hjust=0.5, col="#808080", fontface="bold" )+
  scale_fill_viridis_d()+
  theme_void()+
  theme(legend.position = "none")

```



```
drinks<- read_csv("../drinks.csv")
```

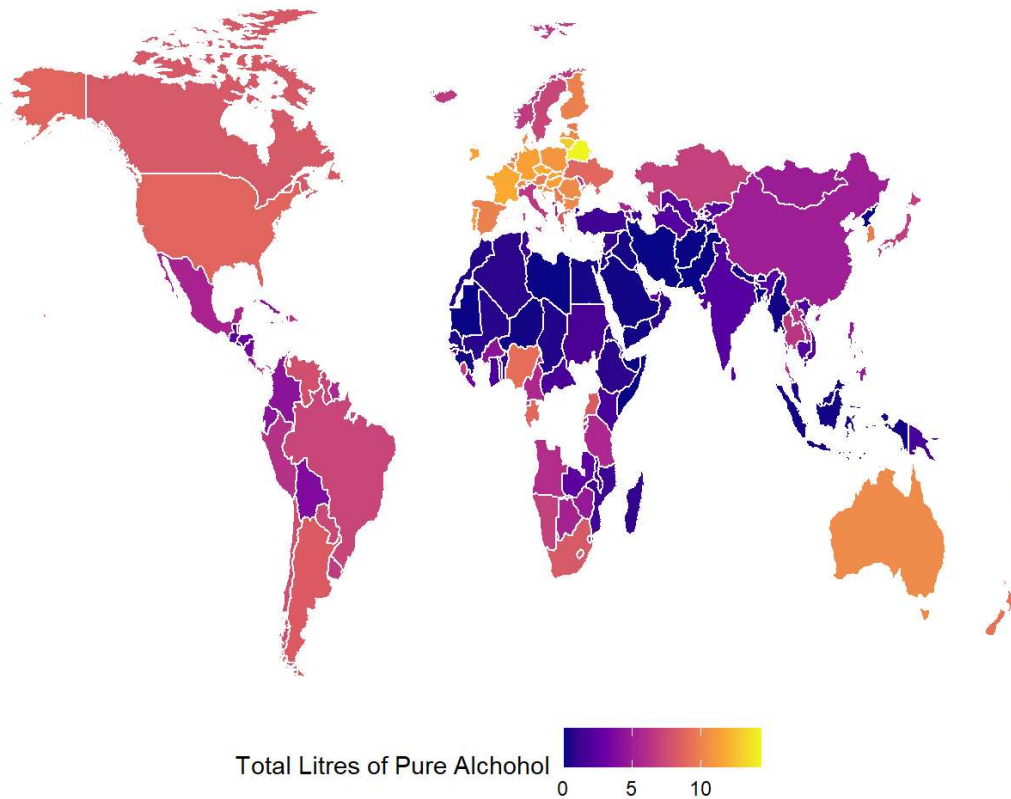
```

## Rows: 193 Columns: 5
## — Column specification —————
## Delimiter: ","
## chr (1): country
## dbl (4): beer_servings, spirit_servings, wine_servings, total_litres_of_pure...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

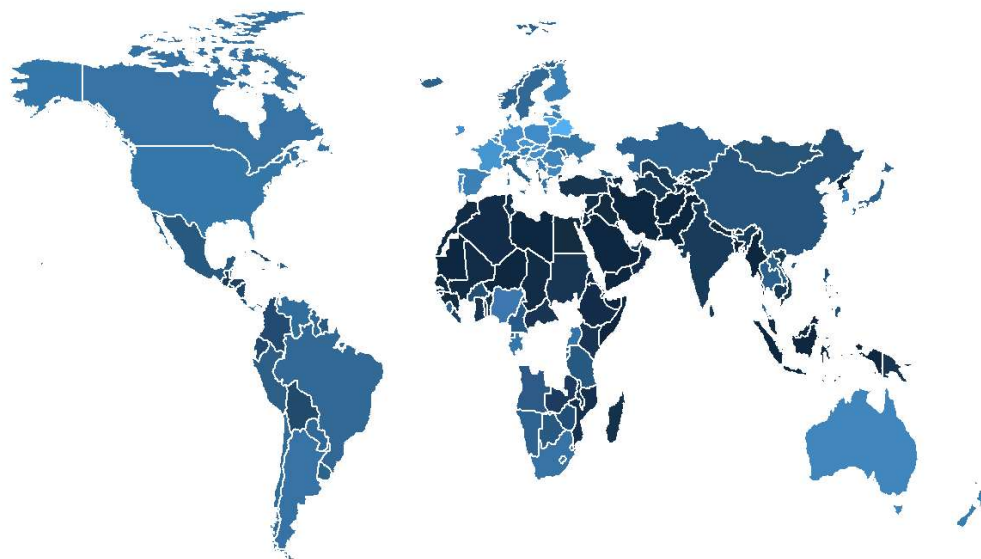
```

```
drinks_map<- drinks%>% left_join(map, by= c("country"= "region"))

ggplot(drinks_map, aes(long, lat, group=group))+
  geom_polygon(aes(fill= total_litres_of_pure_alcohol), colour="white")+
  scale_fill_viridis_c(option = "C")+
  labs(fill="Total Litres of Pure Alchohol")+
  theme_void()+
  theme(legend.position = "bottom")
```



```
ggplot(drinks, aes(map_id=country))+
  geom_map(aes(fill=total_litres_of_pure_alcohol), map = map, colour="white")+
  expand_limits(x=map$long, y= map$lat)+
  labs(fill="Total Litres of Pure Alcohol")+
  theme_void()+
  theme(legend.position = "bottom")
```

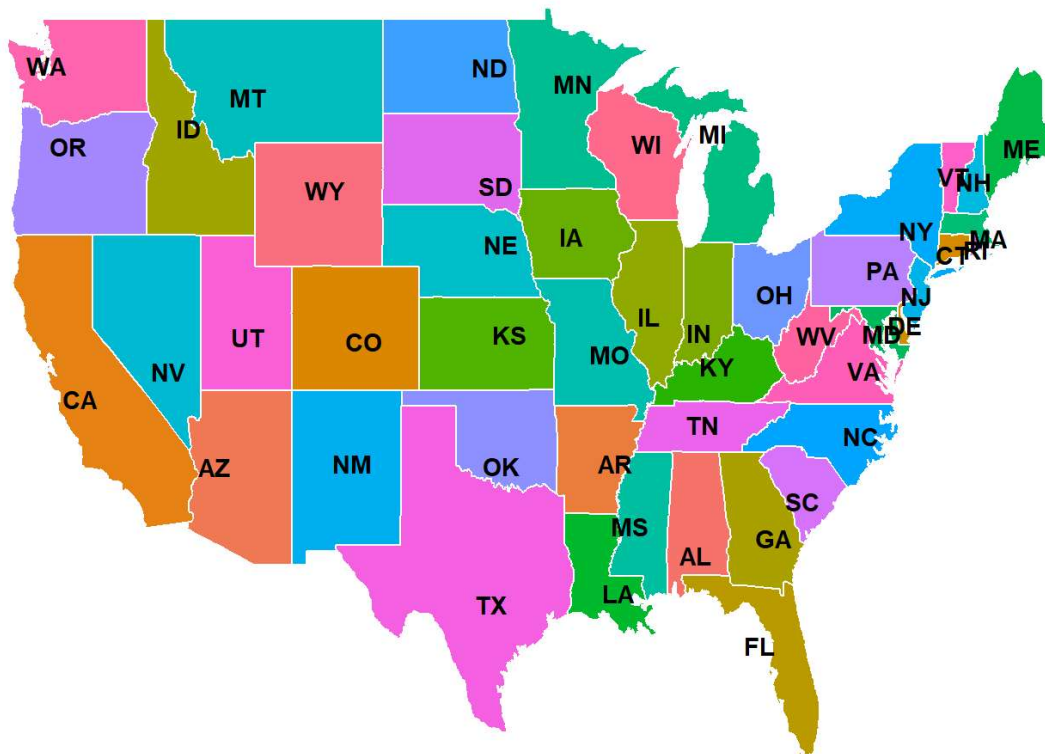


```
US_map<- map_data("state")

state_data<- US_map%%>%
  filter(region != "district of columbia") %>%
  group_by(region)%>%
  summarise(long= mean(long), lat=mean(lat)) %>%
  arrange(region)

state_data$region.abb<- state.abb[-c(2,11)]

p<- ggplot(US_map, aes(x=long, y=lat))+
  geom_polygon(aes(group = group, fill=region), colour="white")+
  geom_text(data=state_data, aes(label = region.abb), fontface="bold")+
  theme_void()+
  theme(legend.position = "none")
p
```

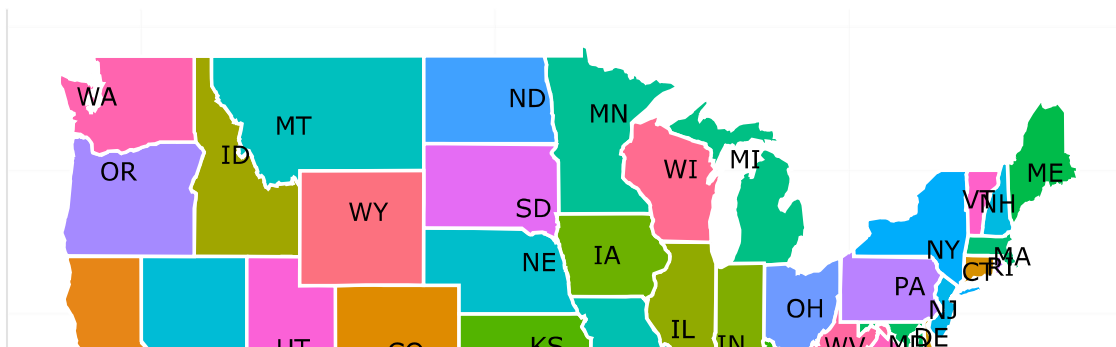


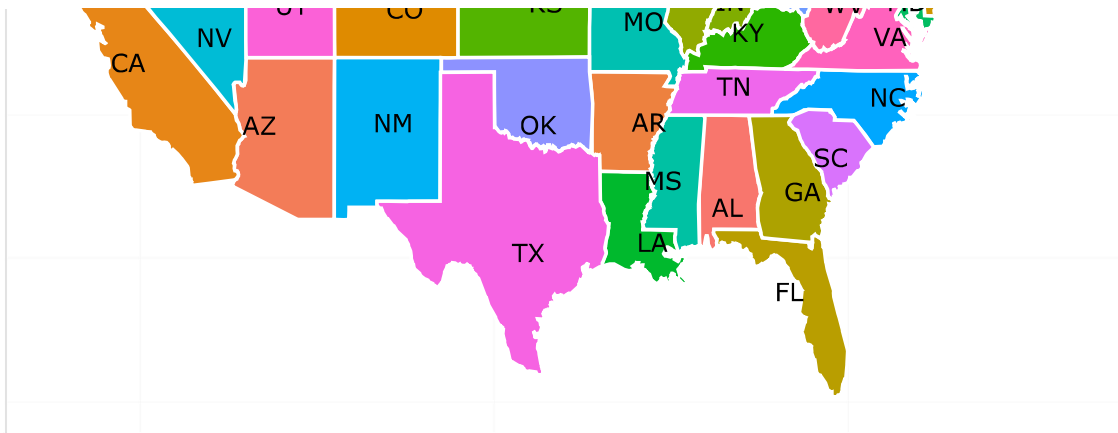
```
library(plotly)
```

```
## Warning: package 'plotly' was built under R version 4.4.3
```

```
##
## Attaching package: 'plotly'
##
## The following object is masked from 'package:ggplot2':
##
##   last_plot
##
## The following object is masked from 'package:stats':
##
##   filter
##
## The following object is masked from 'package:graphics':
##
##   layout
```

```
ggplotly(p)
```





```
US_map<- map_data("state")
state_data<- US_map%>%
  filter(region != "district of columbia") %>%
  group_by(region)%>%
  summarise(long= mean(long), lat=mean(lat)) %>%
  arrange(region)

state_data$region.abb<- state.abb[-c(2,11)]

crimes<- data.frame(region= rownames(USArrests), USArrests)%>%
  filter(region!=c("Alaska", "Hawaii"))

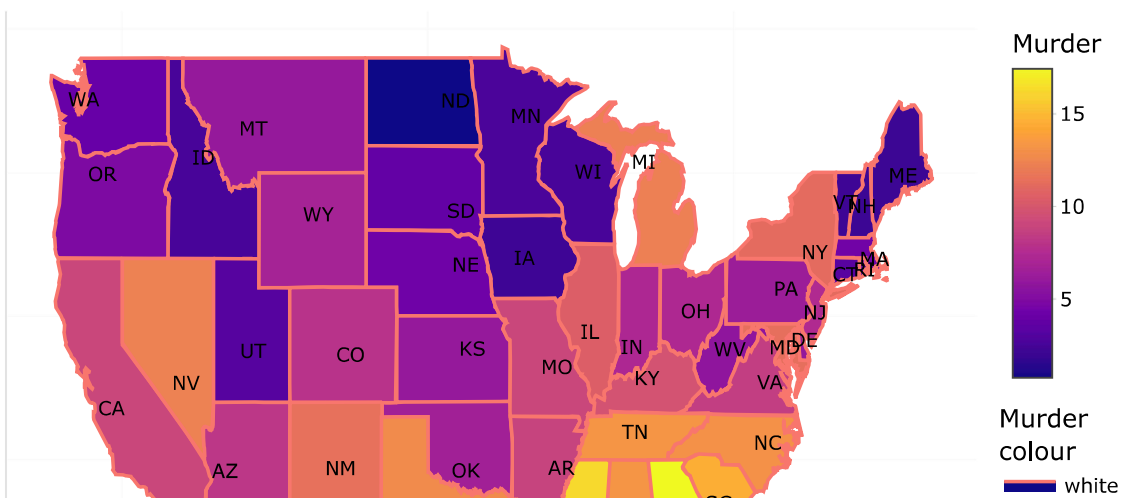
crimes$region<- tolower(crimes$region)

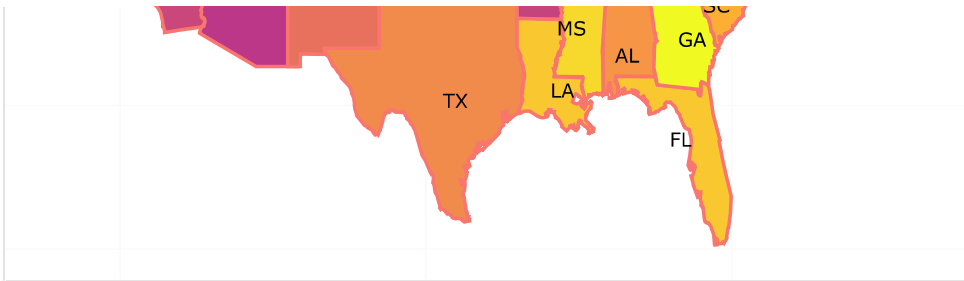
crimes_map<- crimes %>% left_join(US_map, by="region")

g1<- ggplot(crimes_map, aes(x=long, y=lat))+
  geom_polygon(aes(group = group, fill=Murder, text=paste0(region, ":\n", Murder, "murder arrests per 100,000"),
  colour="white"))+
  geom_text(data = state_data, aes(label=region.abb), fontface="bold", size=3)+
  scale_fill_viridis_c(option = "C") +
  theme_void()
```

```
## Warning in geom_polygon(aes(group = group, fill = Murder, text = paste0(region,
## : Ignoring unknown aesthetics: text
```

```
ggplotly(g1, tooltip = "text")
```





```
covid<- read_csv("./COVID19.csv")
```

```
## Rows: 1175665 Columns: 30
## — Column specification —————
## Delimiter: ","
## chr   (4): id, administrative_area_level_1, administrative_area_level_2, adm...
## dbl  (25): confirmed, deaths, people_vaccinated, people_fully_vaccinated, sc...
## date  (1): date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ohio <- covid %>% filter(administrative_area_level_2=="Ohio", date== "2021-12-31")

ohio <- ohio%>%rename(county= administrative_area_level_3 )

ohio_county<- map_data("county", region="ohio")

ohio_county$subregion<- str_to_title(ohio_county$subregion)

ohio_map<- ohio%>% left_join(ohio_county, by= c("county"="subregion"))

g2 <- ggplot(ohio_map, aes(x=long, y= lat))+
  geom_polygon(aes(group=group, fill= deaths, text=paste0("County: ", county, "\n", "Total Deaths: ", deaths)))+
  geom_text(data = ohio, aes(x=longitude, y=latitude, label = county),
            color="white", fontface= "bold")+
  scale_fill_viridis_c(option = "H")+
  theme_minimal()+
  theme_void()
```

```
## Warning in geom_polygon(aes(group = group, fill = deaths, text =
## paste0("County: ", : Ignoring unknown aesthetics: text
```

```
font<- list(family="Arial", size=15, color="white")

label<- list(bgcolor="#232F34", bordercolor= "transparent", font=font )

ggplotly(g2, tooltip = "text", width=850, height = 800)%>%style(hoverlabel=label)%>%
  layout(font=font)
```



